INCH-POUND

A-A-51281A
19 March 1993

SUPERSEDING
A-A-51281
19 Dec 1985

COMMERCIAL ITEM DESCRIPTION

TAPE, SEALING, STERILIZATION INDICATOR (GAS)

The General Services Administration has authorized the use of this Commercial Item Description

This Commercial Item Description covers sterilization indicator tapes suitable for use in ethylene oxide gas sterilization.

Salient characteristics:

The sterilization indicator and sealing tape shall be a creped paper-back design with one side adhesive coated and paper-backed side treated with indicator lines which change color when exposed to ethylene oxide sterilization. The tape shall be wound with the adhesive side in and on a 3 inch (7.62 cm) core to permit dispensing of required lengths by use of a standard tape dispenser. The tape shall be suitable for use in sealing or securing a variety of wraps used to package items for sterilization to include woven, treated woven, nonwoven, paper, plastic and paper/plastic combination wraps. The tape shall also be suitable for application to stainless steel, enamel, porcelain or glass. The tape shall require no moisture, heat or other special preparation prior to use.

AMSC N/A [illegible] FSC 8135

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited

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The adhesive material shall be water-insoluble and allow the tape to be unwound from the roll without separation from the backing. The adhesive material shall adhere immediately and firmly to clean and dry surfaces to which the tape is applied without wrinkling, curling, breaking or lifting, before, during and after sterilization. After sterilization, and when used with reusable woven and treated woven wraps, the adhesive material shall permit removal without breaking and leaving residue on the wraps.

The paper backing shall be impregnated with chemical indicators in the form of lines which shall uniformly and clearly change color to indicate exposure to ethylene oxide sterilization. Pencil or permanent ink marking on the paper backing shall remain legible after sterilization. In addition, the paper backing shall accept adhesive labels such as those normally used to designate expiration dates and sterilizer load records.

The tape shall be provided in the following sizes and styles:

Size 1 - 1/2 inch wide, 60 yards long
Style A - Unit quantity - 36 rolls

Size 2 - 3/4 inch wide, 60 yards long.
Style B - Unit quantity - 48 rolls
Style C - Unit quantity - 24 rolls

Size 3 - 1 inch wide, 60 yards long.
Style D - Unit quantity - 18 rolls

Sterilization conditions Minimum conditions for gas sterilization mentioned herein shall be defined as follows.

Temperature	130°F ± 5°F
Humidity	60 +/- 20 percent RH
Time of exposure EtO	48 minutes
Concentration of EtO	12 percent
Pressure in chamber (follow 25" per-vac)	Atmospheric (28-32 in Hg)

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Conditions similar to sterilization. A color change of the tape shall not occur when it is subjected to conditions similar to ethylene oxide sterilization. These conditions shall be defined as follows.

Temperature	130°F ± 5°F
Humidity	60 +/- 20 percent RH
Time	48 minutes
Concentration of Gas in chamber	Air N2, Freon or other inert gas
Pressure in test chamber	Atmospheric (28-32 in Hg)

Physical properties. The following physical properties shall apply only after conditioning the tape for 24 hours at 73°F ± 3.5°F and 48 percent minimum relative humidity, and with a sample taken after the first three plies are removed from the roll:

Property	Requirement	Test
Thickness, inches	0.011, max.	ASTM D-3652 except average of 5 tests.
Tensile strength, dry, pounds per inch of width	12, min.	ASTM D-3759
Ultimate, elongation, percent	6, min.	ASTM D-3759
Adhesion to steel, ounces per inch of width	20, min.	ASTM D-3330
Color change adhesive	Positive, easily distinguishable	5 mil strip of tape, side applied to stainless steel panel and exposed to herein defined sterilization conditions

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Property	Requirement	Test
Color change	None**	Same as above, except exposed to herein defined conditions similar to sterilization

* Above specified condition of tape shall not apply.

** A slight color change due to dehydration shall be acceptable.

Workmanship. The tape shall be free from defects which detract from its appearance or impair its serviceability. The adhesive shall be a smooth, uniform coating, covering the entire area of one side of the tape. The rolls shall be free of foreign matter and the edges shall be straight.

Unit. Package (PG).

Size 2, Style B. One package containing 48 rolls of tape, as specified, constitutes one unit.

Size 2, Style C. One package containing 24 rolls of tape, as specified, constitutes one unit.

Size 1, Style A. One package containing 36 rolls of tape, as specified, constitutes one unit

Size 3, Style D. One package containing 18 rolls of tape, as specified, constitutes one unit

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QUALITY ASSURANCE PROVISIONS.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination and tests. Sampling for examination and tests shall be conducted in accordance with MIL-STD-105, with an AQL of 2.0 (percent defective) and an inspection level of II. The unit of product for examination purposes shall be one roll of tape.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

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Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor Certification The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug, and Cosmetic Act If the product covered by this document has been determined by the U.S Food and Drug Administration to be under its jurisdiction , the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug, and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered material. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

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Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities, labeling and marking must be specified in the contract and/or order.

NOTES: The following National Stock Numbers are covered by this document:

NSN	Item Identification	Size	Style
6530-01-322-8536	TAPE, SEALING, STERILIZATION INDICATOR (GAS), 48s.	2	B
6530-00-334-4849	TAPE, SEALING, STERILIZATION INDICATOR (GAS), 24s.	2	C
6530-00-068-5006	TAPE, SEALING, STERILIZATION INDICATOR (GAS), 36s.	1	A
6530-01-304-0987	TAPE, SEALING, STERILIZATION INDICATOR (GAS), 18s.	3	D

MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force-03

PREPARING ACTIVITY

DoD-MB

Agent

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
USPHS
EPA-MP[illegible]

Project No. 6530-2396

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER: A-A-51281A	2 DOCUMENT DATE (YYMMDD): 19 March 1993

3 DOCUMENT TITLE: TAPE, SEALING, STERILIZATION INDICATOR (GAS)

4 NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5 REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8 PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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